

Homework

Basics in linear algebra and numerical methods, 2026. semester I.

General instructions: Each week, you have to submit the solution of three arbitrary exercises. This should be done as a Python code. Comments should be written in English.

Scores: 3×3 per week; altogether: 54 in the whole semester.

Week 6

- (a) Choose the parameter α such that the first (simplest) gradient method in `lec10.ipynb` converges to the true minimum. Plot $|\Phi(\mathbf{s}^n) - \Phi(\mathbf{s}^{\text{opt}})|$ as a function of n . Compare this with the function $1/n$. What happens after a certain number of iterations?
- (b) Check the convergence rate of the best (last) method gradient method in `lec10.ipynb` (with the sophisticated learning rate) corresponding to the theoretical result of proportionality between the error and 1 over the number of steps. For a meaningful learning rate α , estimate the exponent r such that $|\Phi(\mathbf{s}^n) - \Phi(\mathbf{s}^{\text{opt}})|$ is proportional to n^{-r} . (You may take the logarithm of the data and find a linear fit using `np.polyfit`). Then compare, using a log log plot of the error as a function of n , with a log log plot of n^{-r} as a function of n . What happens after some iterations?
- (c) For the last code in `lec11.ipynb` take the average of the last 100 approximations to have an estimate for the desired parameters. Also, change inner loop for j to a simple vectorized operation.